Day 078 | Unit 06 | Teacher plan

Understand and Use Ratios and Rates

Compare Ratios Fairly

Standards: 6.RP.A.3a

Learning target: I can compare ratios using a common quantity.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Scale two ratios until one quantity matches, then compare the other quantity. The larger count alone is not enough: compare the amount per equal-sized batch or per one item.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 078 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Compare Ratios Fairly

Standards: 6.RP.A.3a

Learning target: I can compare ratios using a common quantity.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Scale two ratios until one quantity matches, then compare the other quantity. The larger count alone is not enough: compare the amount per equal-sized batch or per one item.

Worked example 1

Mix A has 2 cups juice for 6 cups water. Mix B has 7 cups juice for 18 cups water. Which is stronger in juice? Show equivalent ratios.

1. Scale two ratios until one quantity matches, then compare the other quantity.

2. The larger count alone is not enough: compare the amount per equal-sized batch or per one item.

3. For 18 cups water, A uses 6 cups juice

4. B uses 7.

5. B has more juice per equal water amount.

6. State the result in the form requested: Mix B.

Answer: Mix B

Worked example 2

Mix A uses 2 cups concentrate per 5 cups water. Mix B uses 9 per 20. Select all true statements.
A. The mixtures have equivalent ratios.
B. Mix B is stronger.
C. For 20 cups water, A uses 8 cups concentrate.
D. The larger water amount alone proves A is stronger.

1. Scale two ratios until one quantity matches, then compare the other quantity.

2. Scale A by 4 to the same water amount.

3. B has 1 more cup of concentrate with equal water, so B is stronger.

Answer: B, C - Mix B is stronger.; For 20 cups water, A uses 8 cups concentrate.

Day 078 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Mix A has 5 cups juice for 7 cups water. Mix B has 26 cups juice for 35 cups water. Which is stronger in juice? Show equivalent ratios.

02 / Constructed response

Mix A has 5 cups juice for 2 cups water. Mix B has 33 cups juice for 12 cups water. Which is stronger in juice? Show equivalent ratios.

03 / Constructed response

Mix A has 7 cups juice for 4 cups water. Mix B has 45 cups juice for 24 cups water. Which is stronger in juice? Show equivalent ratios.

04 / Constructed response

Mix A has 6 cups juice for 6 cups water. Mix B has 39 cups juice for 36 cups water. Which is stronger in juice? Show equivalent ratios.

05 / Constructed response

Mix A has 7 cups juice for 6 cups water. Mix B has 38 cups juice for 30 cups water. Which is stronger in juice? Show equivalent ratios.

06 / Constructed response

Mix A has 6 cups juice for 2 cups water. Mix B has 20 cups juice for 6 cups water. Which is stronger in juice? Show equivalent ratios.

07 / Constructed response

Mix A has 7 cups juice for 5 cups water. Mix B has 44 cups juice for 30 cups water. Which is stronger in juice? Show equivalent ratios.

08 / Multiple select

Mix A uses 5 cups concentrate per 5 cups water. Mix B uses 26 per 25. Select all true statements.
A. For 25 cups water, A uses 25 cups concentrate.
B. The larger water amount alone proves A is stronger.
C. Mix B is stronger.
D. The mixtures have equivalent ratios.

09 / Multiple select

Mix A uses 5 cups concentrate per 6 cups water. Mix B uses 26 per 30. Select all true statements.
A. For 30 cups water, A uses 25 cups concentrate.
B. The larger water amount alone proves A is stronger.
C. The mixtures have equivalent ratios.
D. Mix B is stronger.

10 / Constructed response

Mix A has 7 cups juice for 5 cups water. Mix B has 39 cups juice for 25 cups water. Which is stronger in juice? Show equivalent ratios.

Day 078 | Unit 06 | Answer key

Understand and Use Ratios and Rates

Compare Ratios Fairly

Standards: 6.RP.A.3a

Learning target: I can compare ratios using a common quantity.

1. Mix B

For 35 cups water, A uses 25 cups juice; B uses 26. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

2. Mix B

For 12 cups water, A uses 30 cups juice; B uses 33. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

3. Mix B

For 24 cups water, A uses 42 cups juice; B uses 45. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

4. Mix B

For 36 cups water, A uses 36 cups juice; B uses 39. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

5. Mix B

For 30 cups water, A uses 35 cups juice; B uses 38. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

6. Mix B

For 6 cups water, A uses 18 cups juice; B uses 20. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

7. Mix B

For 30 cups water, A uses 42 cups juice; B uses 44. B has more juice per equal water amount.

6.RP.A.3a | Constructed response

8. A, C - For 25 cups water, A uses 25 cups concentrate.; Mix B is stronger.

Scale A by 5 to the same water amount. B has 1 more cup of concentrate with equal water, so B is stronger.

6.RP.A.3a | Multiple select

9. A, D - For 30 cups water, A uses 25 cups concentrate.; Mix B is stronger.

Scale A by 5 to the same water amount. B has 1 more cup of concentrate with equal water, so B is stronger.

6.RP.A.3a | Multiple select

10. Mix B

For 25 cups water, A uses 35 cups juice; B uses 39. B has more juice per equal water amount.

6.RP.A.3a | Constructed response